A judicious combination of CO_2, CFC, and NH_3 greenhouse effects on Mars looks as if it might be able to bring surface temperatures close enough to the freezing point of water for the second phase of Martian terraforming to begin—temperatures rising due to the pressure of substantial water vapor in the air, widespread production of O_2 by genetically engineered plants, and fine-tuning the surface environment. Microbes and larger plants and animals could be established on Mars before the overall environment was suitable for unprotected human settlers.

Terraforming Mars is plainly much easier than terraforming Venus. But it is still very expensive by present standards, and environmentally destructive. If there were sufficient justification, though, perhaps the terraforming of Mars could be under way by the twenty-second century.

THE MOONS OF JUPITER AND SATURN: Terraforming the satellites of the Jovian planets presents varying degrees of difficulty. Perhaps the easiest to contemplate is Titan. It already has an atmosphere, made mainly of N_2 like the Earth's, and is much closer to terrestrial atmospheric pressures than either Venus or Mars. Moreover, important greenhouse gases, such as NH_3 and H_2O, are almost certainly frozen out on its surface. Manufacture of initial greenhouse gases that do not freeze out at present Titan temperatures plus direct warming of the surface by nuclear fusion could, it seems, be the key early steps to one day terraform Titan.

IF THERE WERE A COMPELLING REASON for terraforming other worlds, this greatest of engineering projects might be feasible on the timescale we've been describing—certainly for asteroids, possibly for Mars, Titan, and other moons of the outer planets, and probably not for Venus. Pollack and I recognized that there are those who feel a powerful attraction to the idea of rendering other worlds in the Solar System suitable for human habitation—in establishing observatories,